

Exploring the AKS cluster – Kubernetes side

- **cloud-node-manager**
 - Introduced in AKS by default with Kubernetes version 1.22 as a DaemonSet
 - AKS adopted an “out-of-tree” model where the cloud providers can control their releases independently of the core Kubernetes release schedule. cloud-node-manager is a component of the node lifecycle controller, which, in turn, is part of cloud controller manager. cloud-node-manager works with the node lifecycle controller to:
 - Update the nodes with the system’s unique identifier from the cloud provider

```
ProviderID: azure:///subscriptions/c1e88e2e-f469-4d8e-aaab-c7422285cb36/resourceGroups/mc_aks-rg_aks_eastus/providers/Microsoft.Compute/virtualMachineScaleSets/aks-nodepool1-33156667-vmss/virtualMachines/0
```

- Annotate and label the nodes with cloud specific information, for example, the OS, role, and others

```
kubernetes.azure.com/os-sku=Ubuntu
kubernetes.azure.com/role=agent
```

- Get the node’s hostname and network address

```
$kubectl describe node aks-nodepool1-33156667-vmss000000 | grep "Addresses" -A2
Addresses:
  InternalIP:  10.224.0.4
  Hostname:    aks-nodepool1-33156667-vmss000000
```

- Verify the health of the node